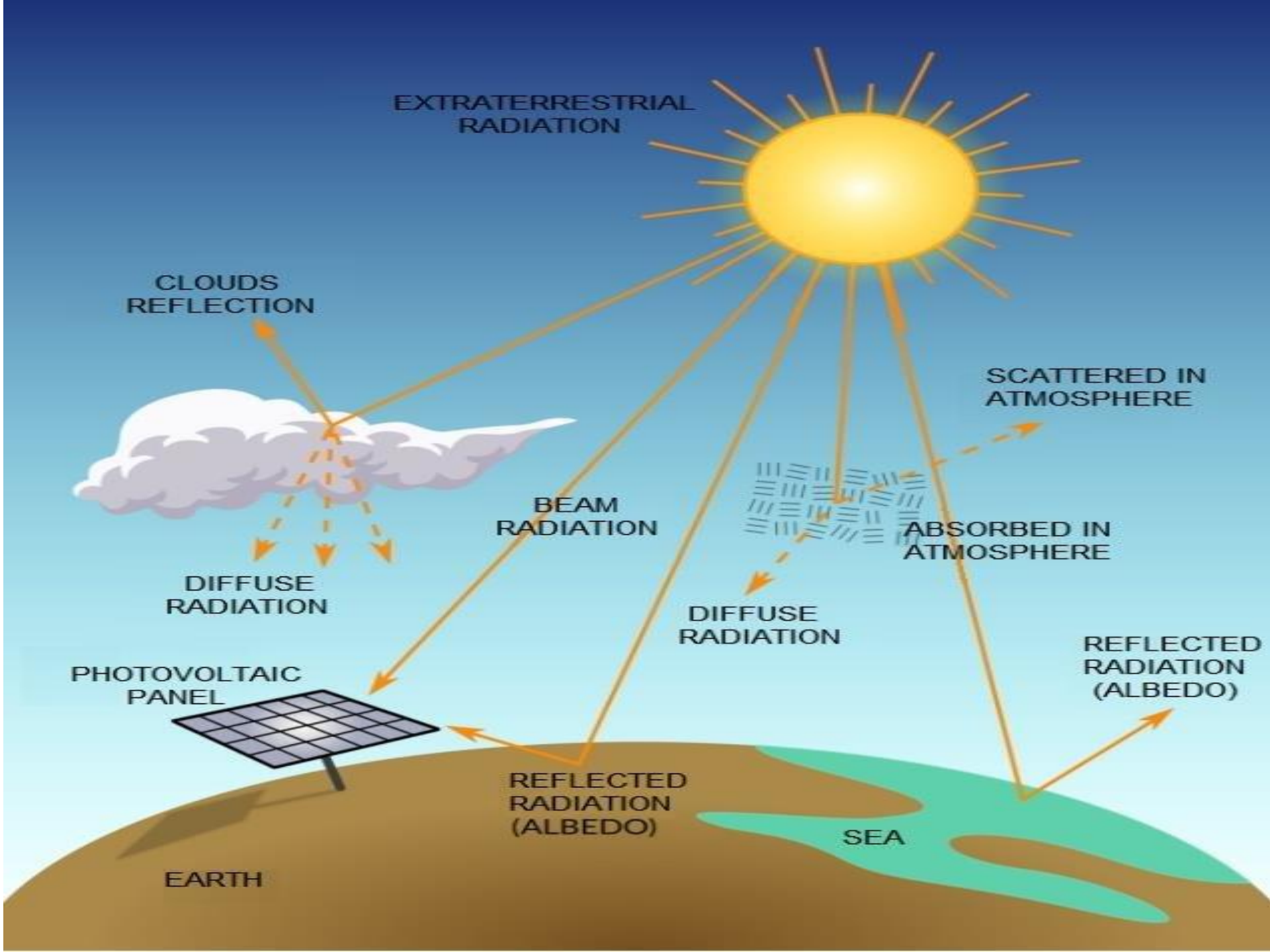
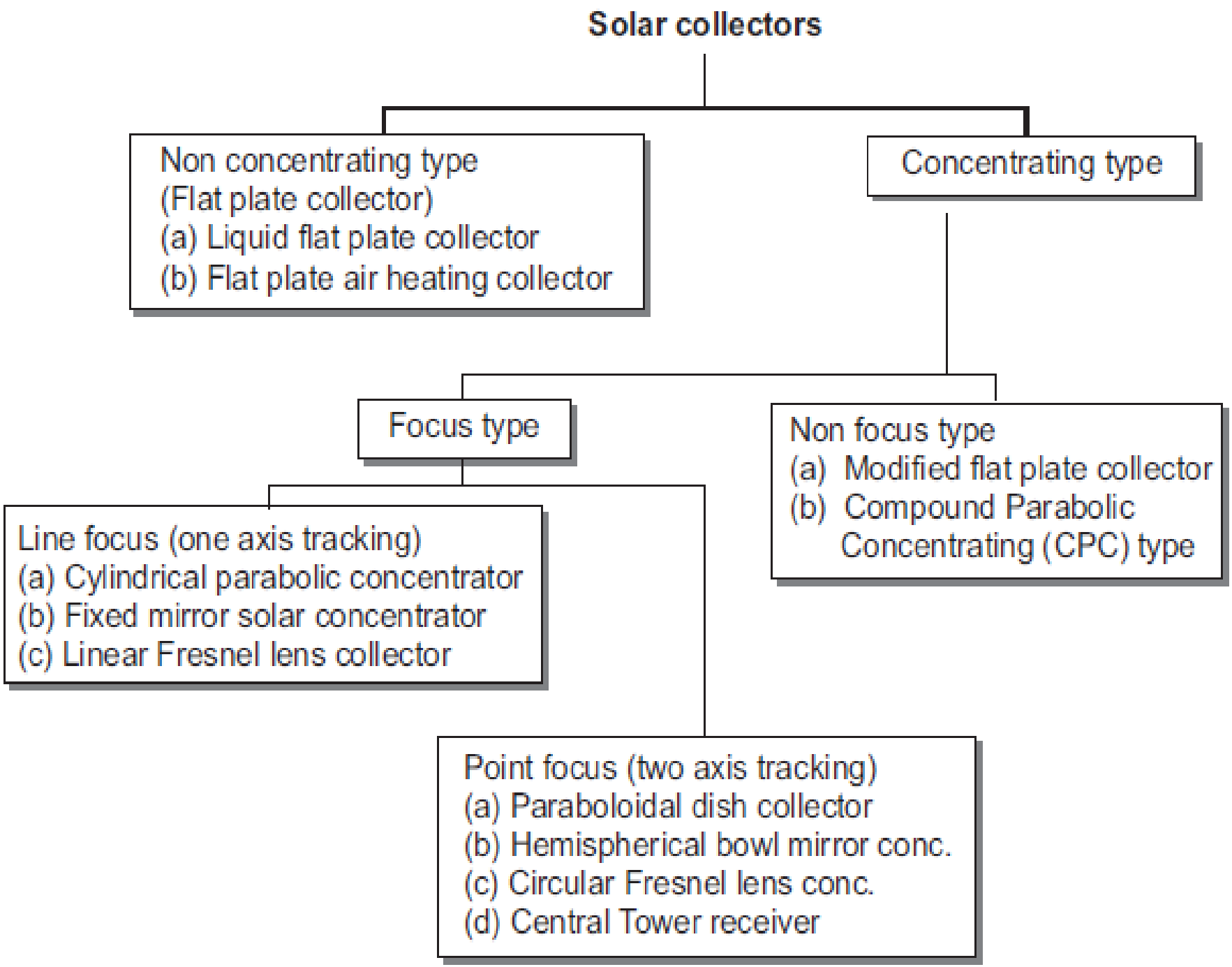
Solar Collectors

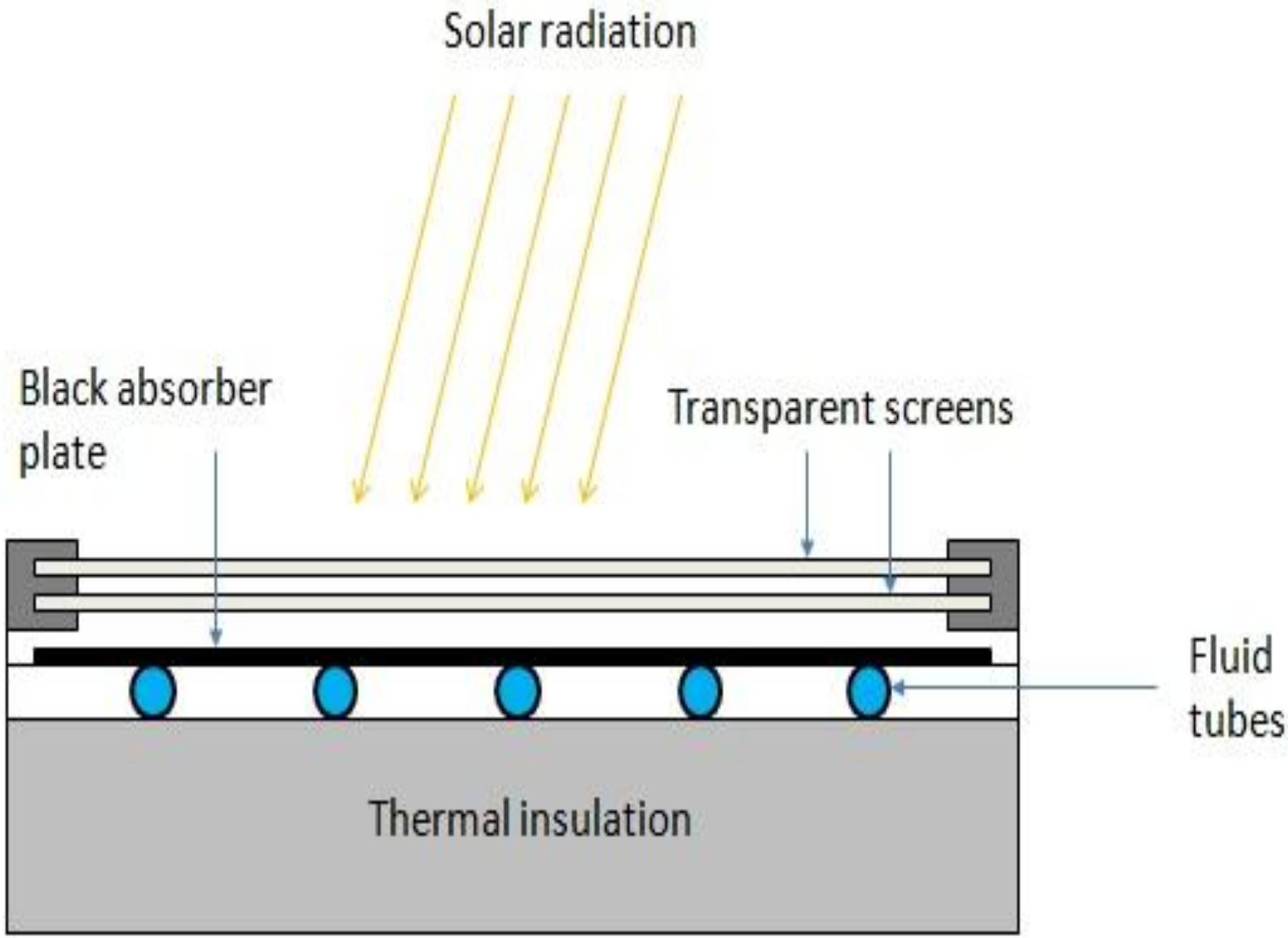

EXTRATERRESTRIAL RADIATION
CLOUDS REFLECTION
SCATTERED IN ATMOSPHERE
BEAM RADIATION
ABSORBED IN ATMOSPHERE
DIFFUSE RADIATION
DIFFUSE RADIATION
REFLECTED RADIATION (ALBEDO)
PHOTOVOLTAIC PANEL
REFLECTED RADIATION (ALBEDO)
SEA
EARTH


Solar collectors
Non concentrating type
(Flat plate collector)
(a) Liquid flat plate collector
(b) Flat plate air heating collector
Concentrating type
Focus type
Non focus type
(a) Modified flat plate collector
(b) Compound Parabolic Concentrating (CPC) type
Line focus (one axis tracking)
(a) Cylindrical parabolic concentrator
(b) Fixed mirror solar concentrator
(c) Linear Fresnel lens collector
Point focus (two axis tracking)
(a) Paraboloidal dish collector
(b) Hemispherical bowl mirror conc.
(c) Circular Fresnel lens conc.
(d) Central Tower receiver

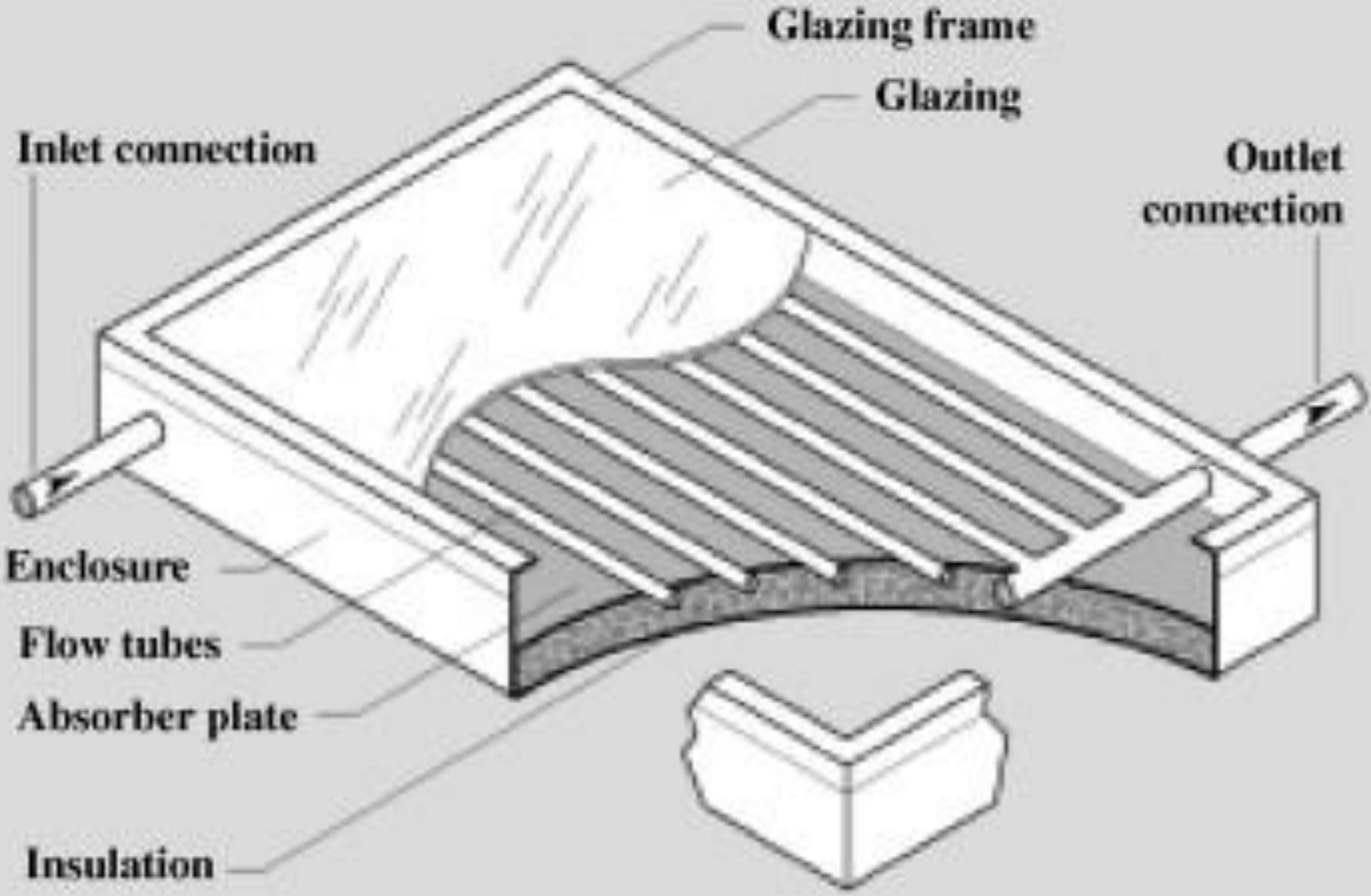
Flat-plate collector

- These are the main components of a typical flat-plate solar collector:
- Black surface - absorbent of the incident solar energy
- Glazing cover - a transparent layer that transmits radiation to the absorber, but prevents radiative and convective heat loss from the surface
- Tubes containing heating fluid to transfer the heat from the collector
- Support structure to protect the components and hold them in place
- Insulation covering sides and bottom of the collector to reduce heat losses

- The flat-plate systems normally operate and reach the maximum efficiency within the temperature range from 30 to 80 ^{o}C ,however, some new types of collectors that employ vacuum insulation can achieve higher temperatures (up to 100 ^{o}C).
- Due to the introduction of selective coatings, the stagnant fluid temperature in flat-plate collectors has been shown to reach 200 ^{o}C.

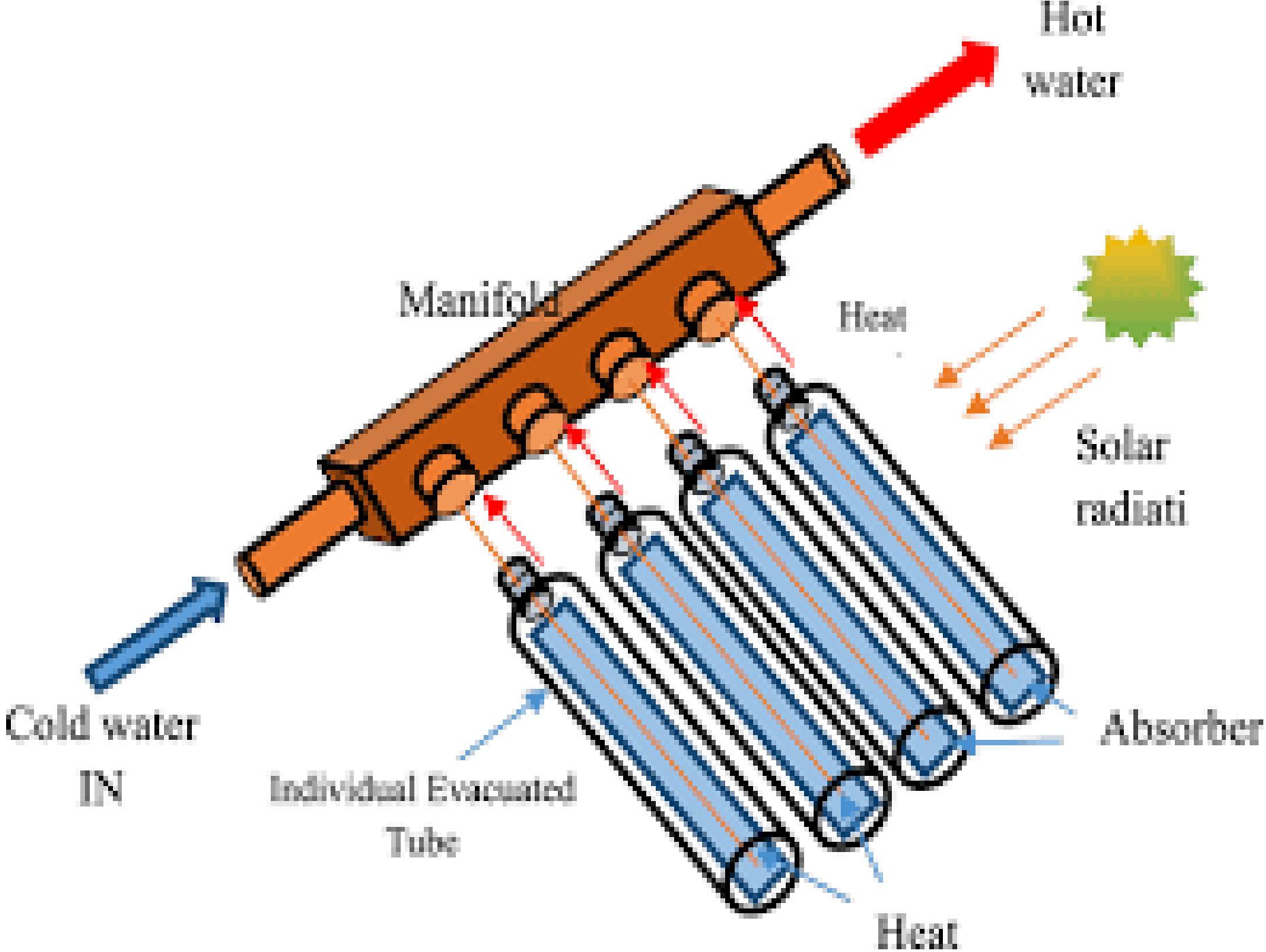

Solar radiation
Black absorber plate
Transparent screens
Fluid tubes
Thermal insulation

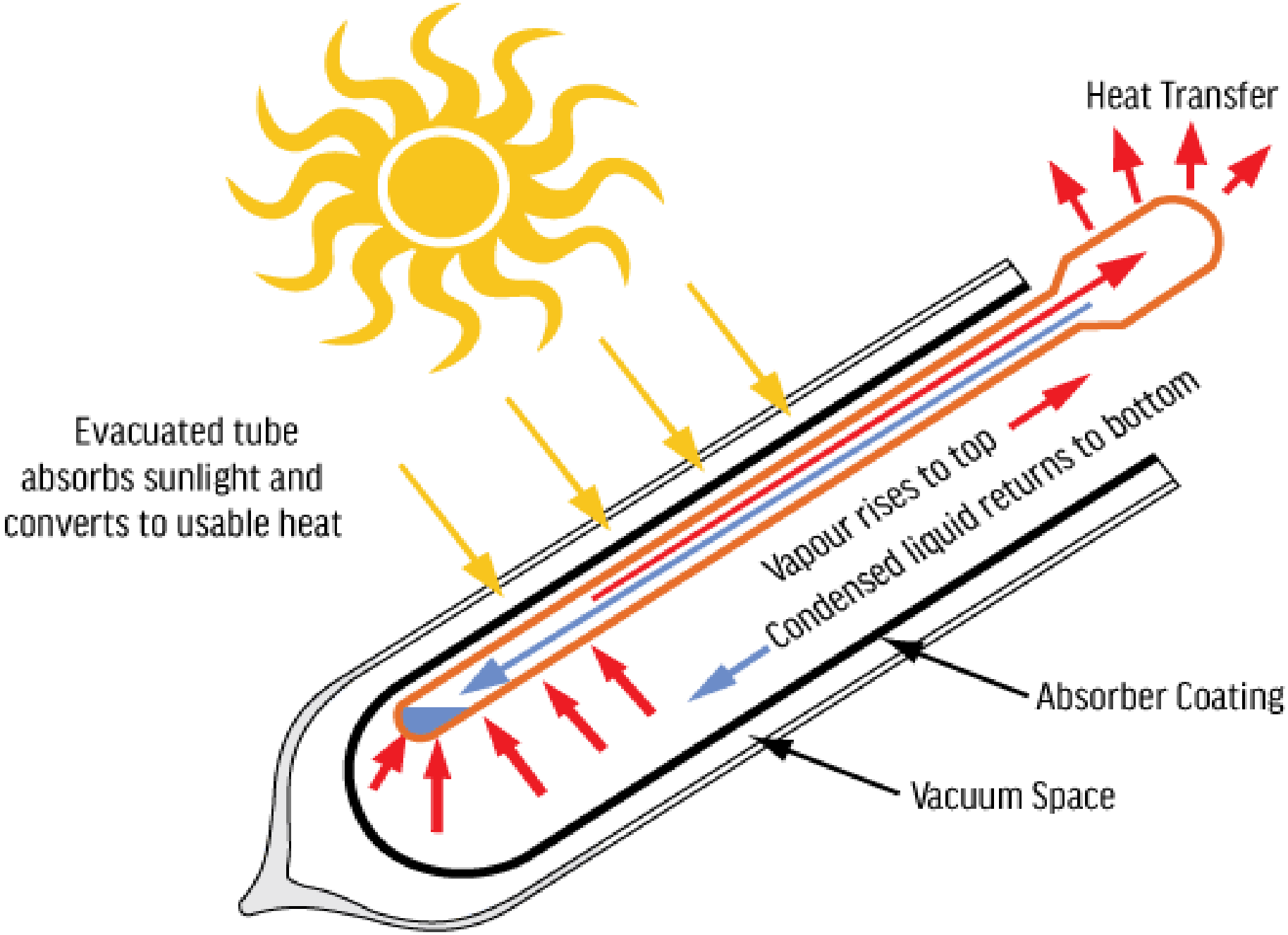
Flat-Plate Collector

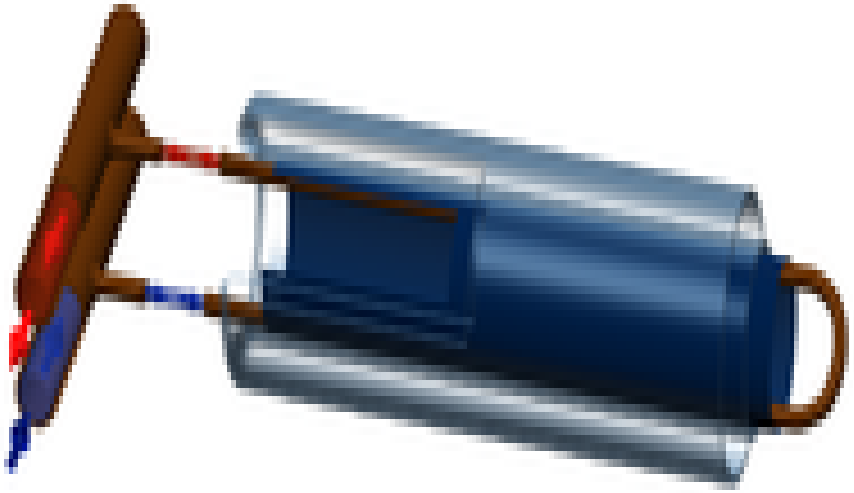


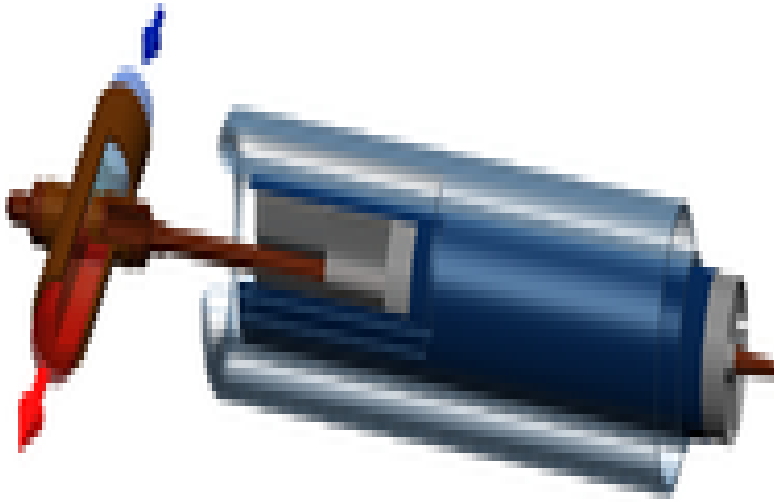
Evacuated-tube solar collectors

- They feature parallel rows of transparent glass tubes.
- Each tube contains a glass outer tube and metal absorber tube attached to a fin.
- The fin's coating absorbs solar energy but inhibits radiative heat loss.
- These collectors are used more frequently for U.S. commercial applications.


Hot
water
Manifold
Heat
Solar
radiati
Cold water
IN
Individual Evacuated
Tube
Absorber
Heat


Heat Transfer
Evacuated tube absorbs sunlight and converts to usable heat
Vapour rises to top
Condensed liquid returns to bottom
Absorber Coating
Vacuum Space





Concentrating collectors

- Concentration of solar radiation is achieved using a reflecting arrangement of mirrors or a refracting arrangement of lenses.
- The optical system will allow the solar radiation to be directed to absorber of smaller area which is usually surrounded by a transparent cover. Because of the optical system, certain losses are introduced.
- These include reflection or absorption losses in the mirrors or lenses and losses due to geometrical imperfections in the optical system.
- Due to the presence of an optical system, a concentrating collector usually has to follow or track the sun so that the beam radiation is directed on to the absorber surface.
- As the solar concentrators mainly makes use of direct beam sun rays and is concentrated high temperatures can be achieved. The method of tracking adopted and the precision with which it has to be done varies considerably.
- In collectors giving a low degree of concentration, it is often adequate to make one or two adjustments of the collector orientation every day. These can be made manually. On the other hand, with collectors giving high degree of concentration, it is necessary to make continuous adjustments of the collector orientation.
- The need for some form of tracking introduces a certain amount of complexity in the design. Maintenance requirements were also increased due to that..

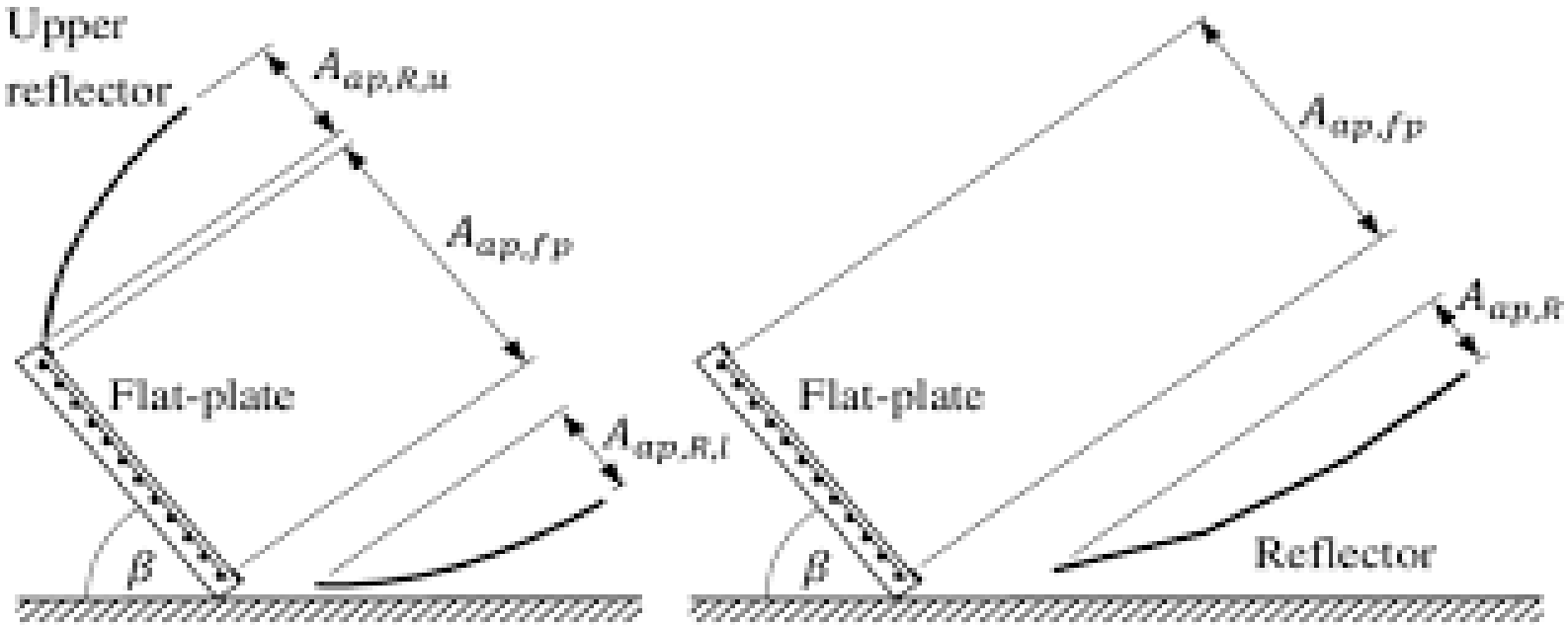
Terms used in solar concentrators

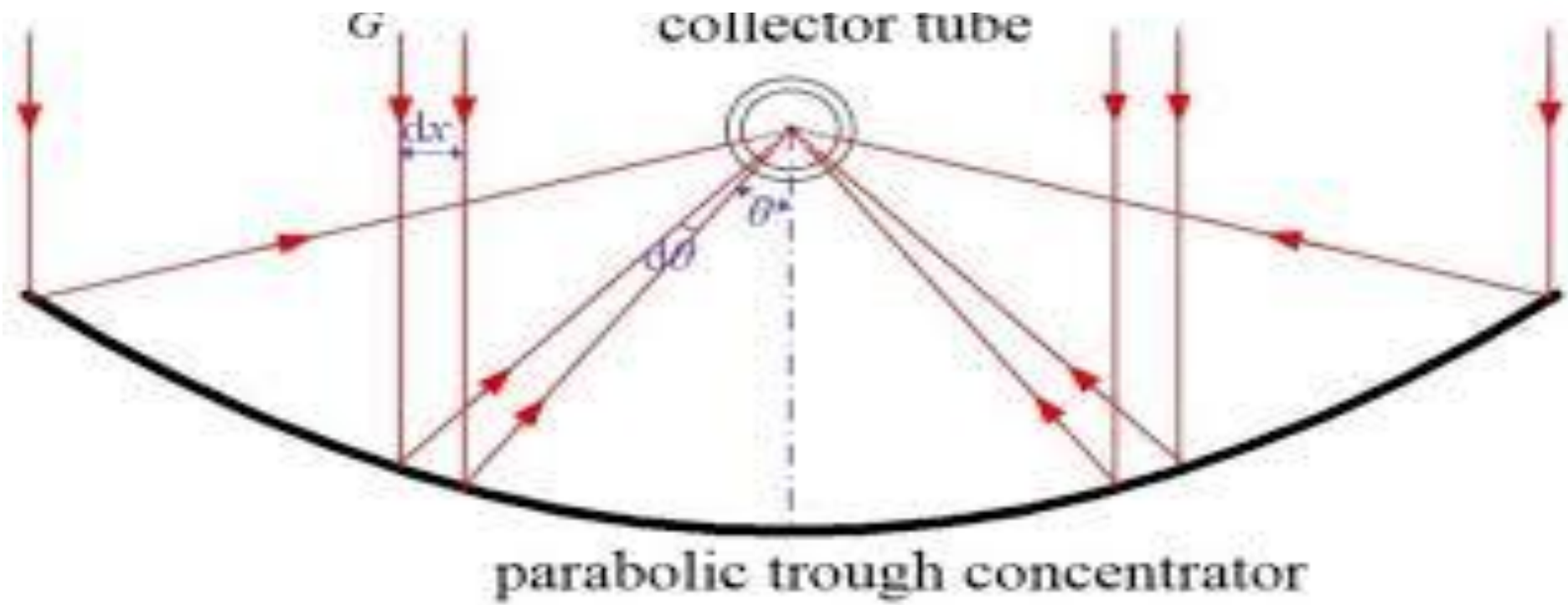
- ‘**Concentrator**’ is for the optical subsystem that projects solar radiation to absorber. Receiver represents a sub system that includes the absorber, its cover and accessories.

- ‘**Aperture**’ (W) is the opening of the concentrator through which solar radiation passes.

- ‘**Acceptance angle**’ (2ø) is the angle across which beam radiation may deviate from the normal to the aperture plane and then reach the absorber.

- ‘**Concentration ratio**’ (CR) is the ratio of the effective area of the aperture to the surface area of the absorber. The value of CR may change from unity to a thousand. The CR is used to classify collectors by their operating temperature range.

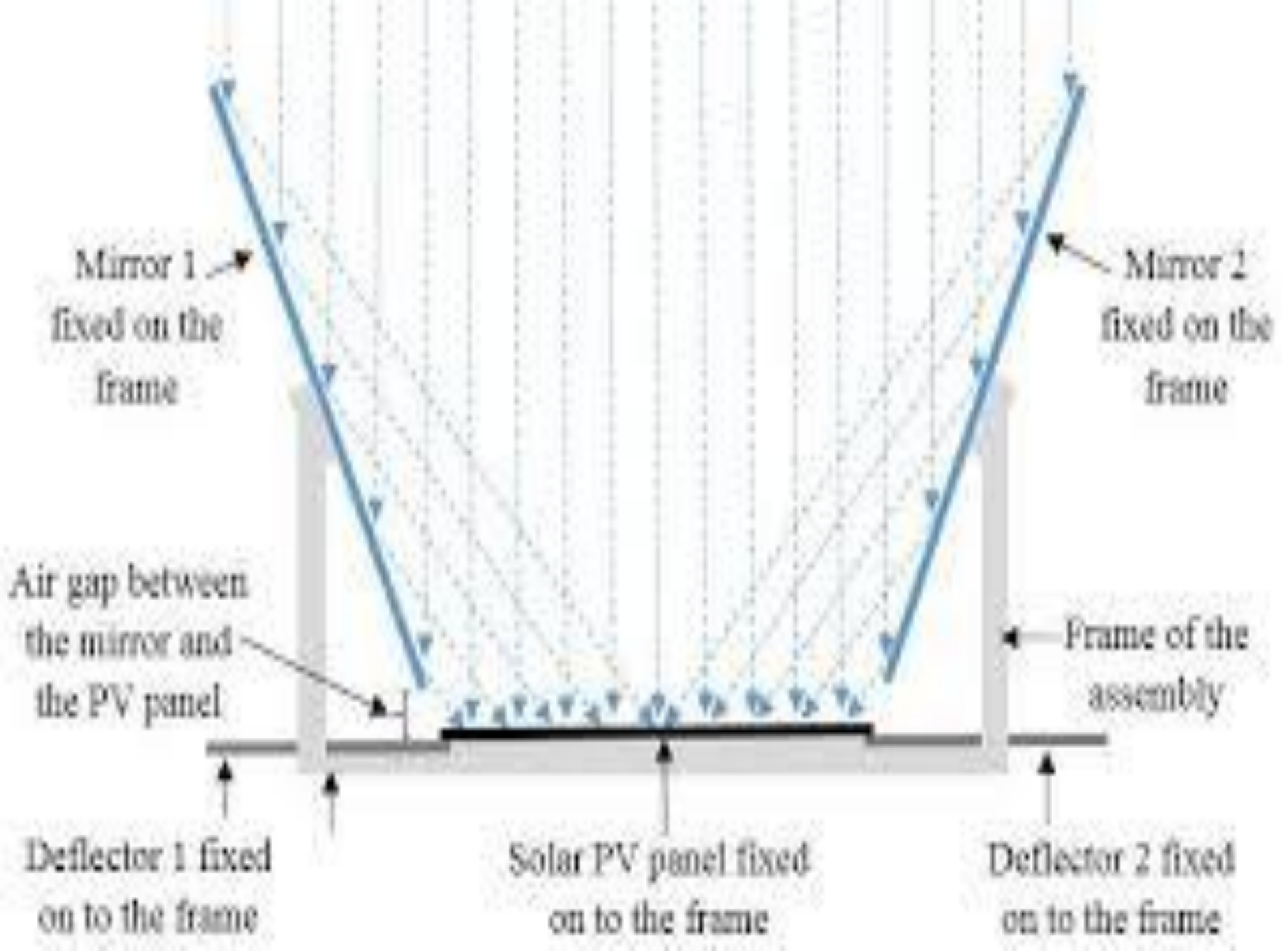

Upper
reflector
$A_{ap,R,u}$
$A_{ap,fp}$
Flat-plate
$A_{ap,R,l}$
β
$A_{ap,fp}$
$A_{ap,R}$
Flat-plate
Reflector
β


G
collector tube
dx
θ
$d\theta$
parabolic trough concentrator

Types of concentrating collectors

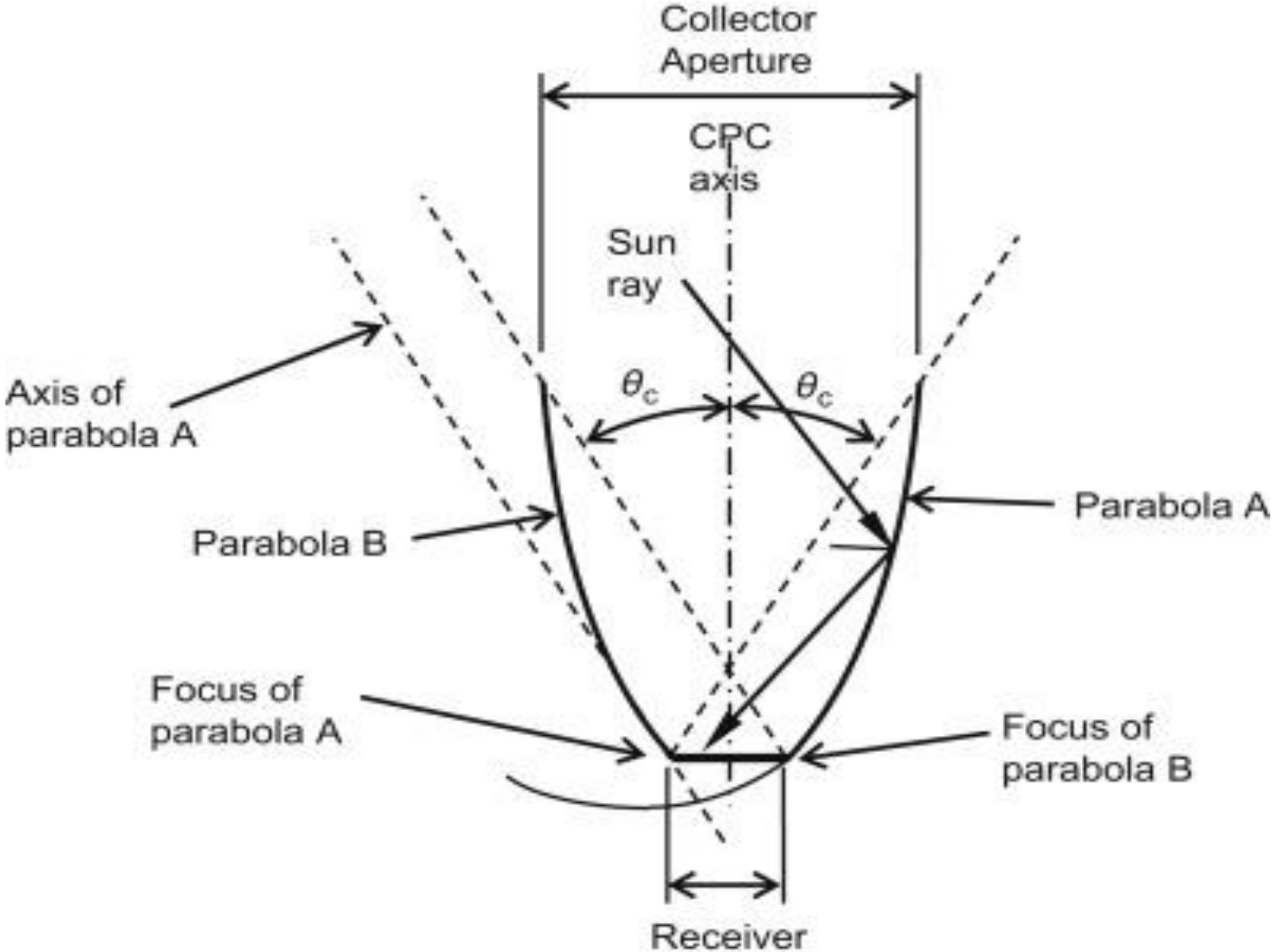
Flat plate collectors with adjustable mirrors or plane reflectors

- A "flat plate collector with adjustable mirrors or plane reflectors" refers to a solar thermal collector design where a standard flat plate absorber is augmented with adjustable mirrors (or plane reflectors) positioned around its edges to direct more sunlight onto the absorber plate, thereby increasing its efficiency by concentrating the solar radiation on a larger area; essentially enhancing the performance of a basic flat plate collector by utilizing reflected sunlight.


Mirror 1 fixed on the frame
Mirror 2 fixed on the frame
Air gap between the mirror and the PV panel
Frame of the assembly
Deflector 1 fixed on to the frame
Solar PV panel fixed on to the frame
Deflector 2 fixed on to the frame

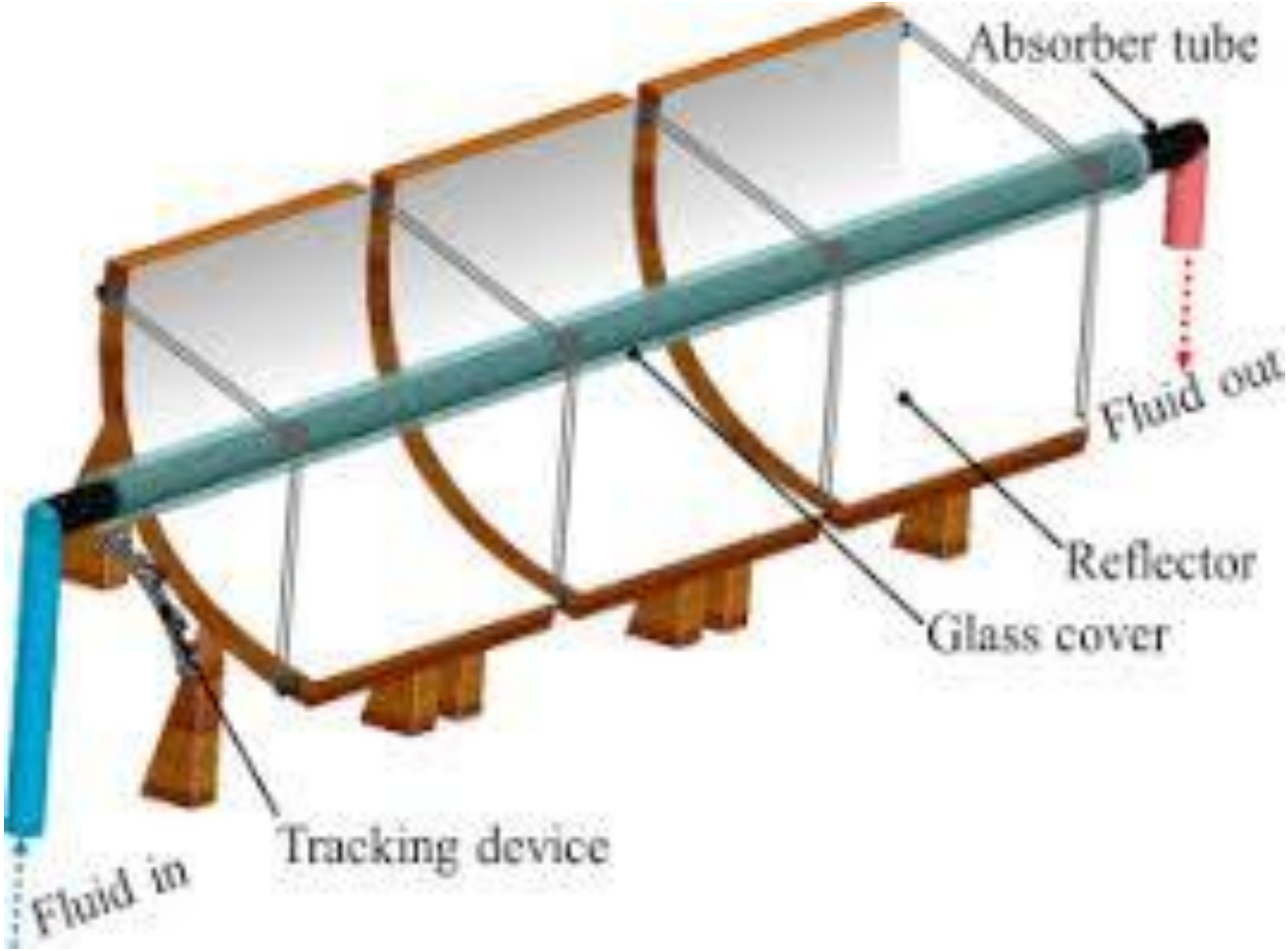
Compound parabolic concentrating collector (CPC)

- A compound parabolic collector (CPC) is type of solar thermal collector that can concentrate and reflect all incident radiation onto a receiver.
- Their potential as collectors of solar energy was highlighted by Winston , therefore sometimes they are called Winston-type collectors.
- CPC is characterized by low concentration (<5) and medium output temperature.
- The reflector geometry is built by the combination of two symmetric parabolic segments with different focus points facing each other.
- This geometry allows CPC to operate without continuous tracking and to receive radiation that is incident on the collector aperture within an acceptance angle .
- CPCs should have a small gap between the absorber and the reflector to eliminate heat loss from the absorber through the reflector.


Collector
Aperture
CPC
axis
Sun
ray
Axis of
parabola A
θ_c
θ_c
Parabola A
Parabola B
Focus of
parabola A
Focus of
parabola B
Receiver

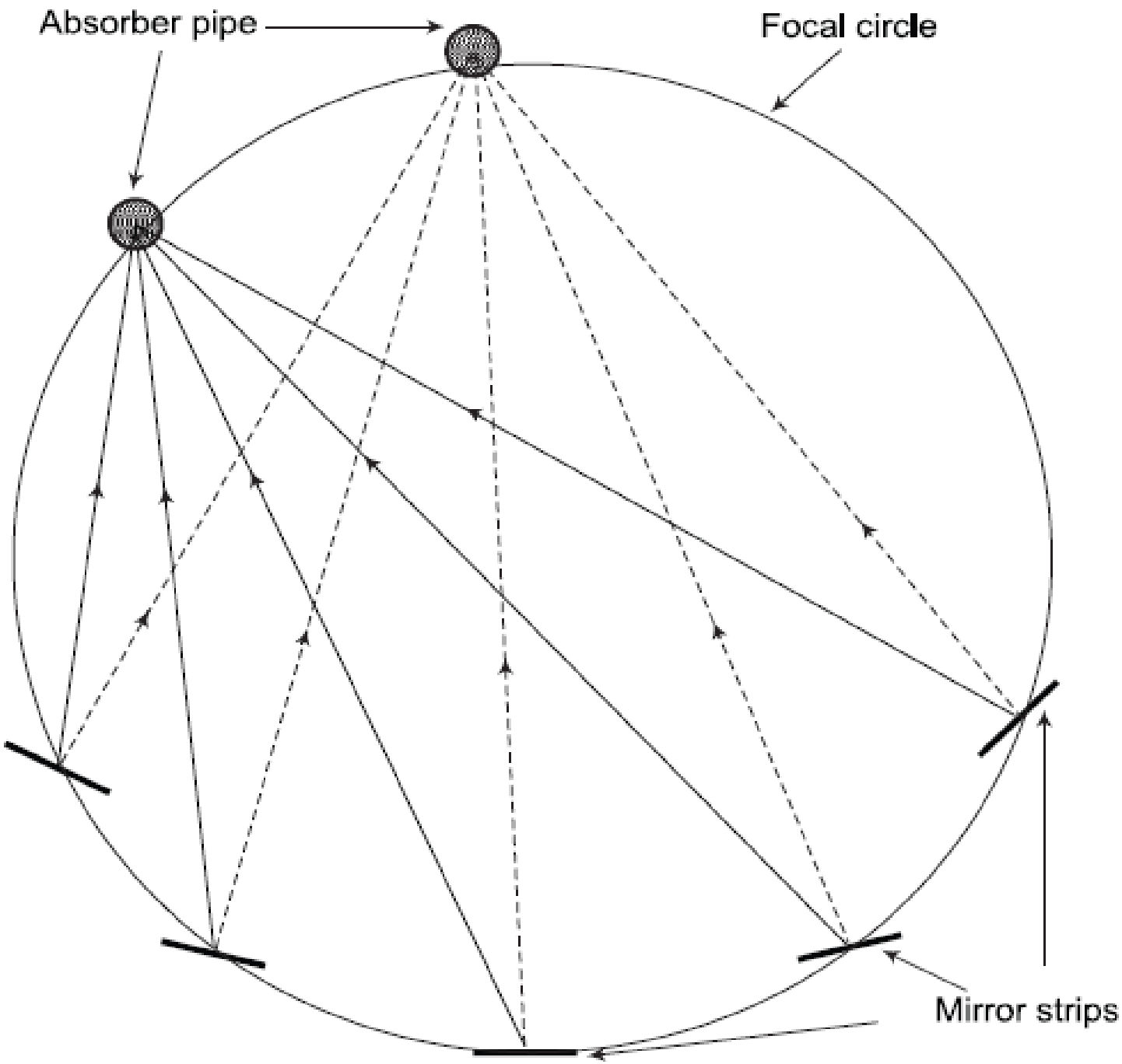
Cylindrical parabolic collector

- A cylindrical parabolic collector is a device that focuses the sun's energy onto a receiver to generate heat.
- It's a type of solar collector that uses parabolic mirrors to reflect sunlight onto a tube or pipe.
- The collector's parabolic mirrors reflect sunlight onto a tube or pipe .The tube or pipe is filled with a heat transfer fluid, like water .
- The collector may have a driving system to track the sun and maintain focus .
- Cylindrical parabolic collectors are efficient and can be used for a variety of temperatures and applications .
- They can be used to generate electricity by powering steam power cycles .
- Cylindrical parabolic collectors can be expensive to maintain.
- They can require large areas of land


Absorber tube
Fluid out
Reflector
Glass cover
Tracking device
Fluid in

Collector with fixed circular concentrator and moving receiver

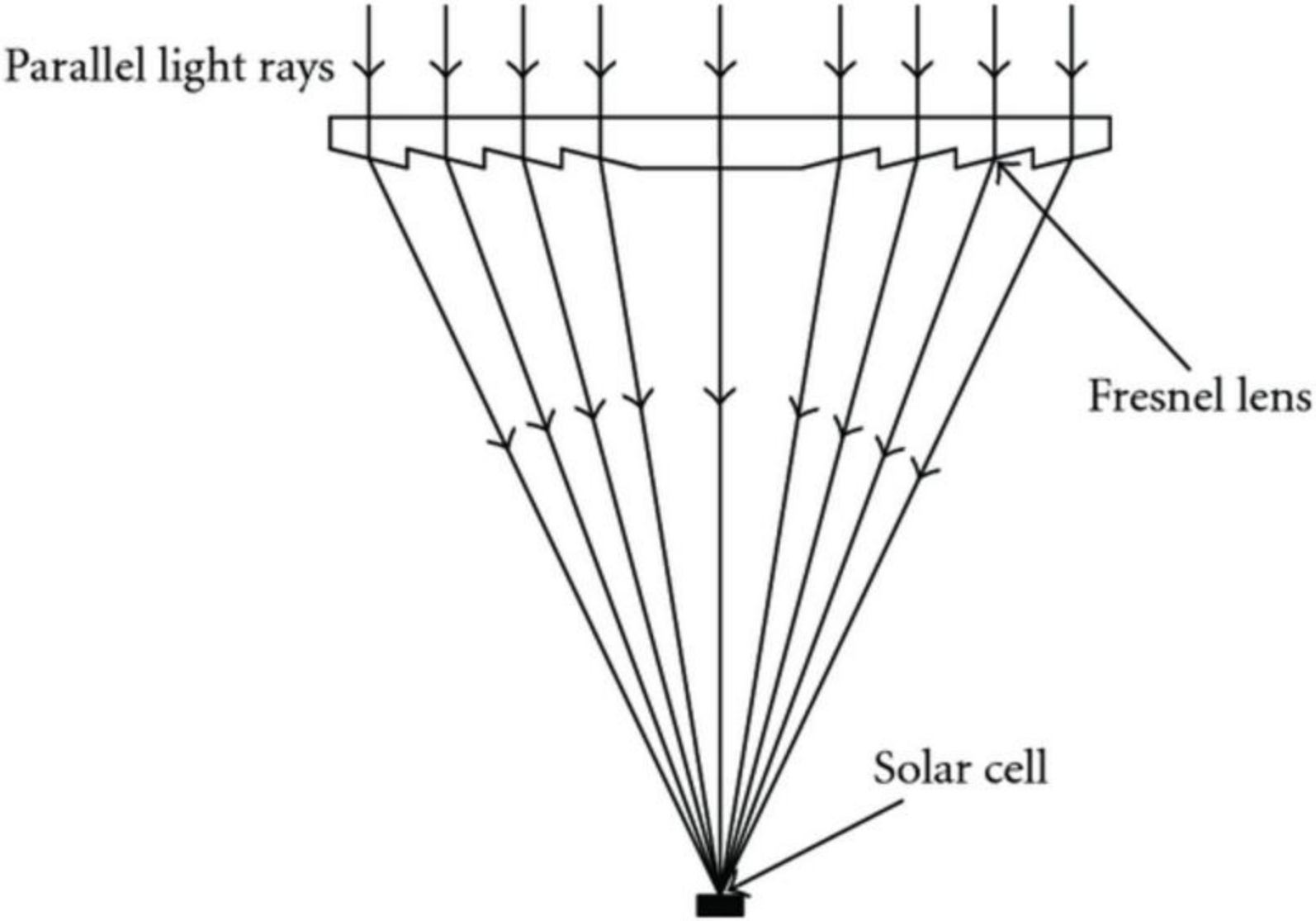
- Collector with a fixed circular concentrator and moving Receiver.
- In this, it has a moving receiver and a fixed concentrator.
- The concentrator is like an array of long and narrow, flat mirror strips fixed along a cylindrical surface.
- In this type of concentrator an array of long, narrow, flat mirror strips fixed along a cylindrical surface.
- The mirror strips produce a narrow line image which follows a circular path as sun moves.
- This path on the same circle on which the mirror strips are fixed.
- Thus receiver has to be moved along the circular path in order to track the sun.


Absorber pipe
Focal circle
Mirror strips

Fresnel lens concentrating collector

- A Fresnel lens concentrating collector is a solar concentrator that uses a Fresnel lens to gather sunlight and focus it onto a receiver.
- The concentrated light can be used for solar power, heating, or cooking.
- The Fresnel lens is made up of segments that concentrate light onto a receiver.
- The lens is designed to focus light efficiently, which increases the concentration ratio. The concentrated light can be used to heat water, generate steam, or power photovoltaic cells.

- **Cost-effective**: Fresnel lenses are cheaper than mirrors.
- **Lightweight**: Fresnel lenses are lightweight and can be used in flexible frames.
- **Efficient**: Fresnel lenses have high optical efficiency and can achieve high concentration ratios.
- **Versatile**: Fresnel lenses can be used for a variety of applications, including solar cooking, industrial heating, and direct steam generation.
- **Solar power**: Fresnel lens collectors can be used to generate electricity through photovoltaic cells.
- **Heating**: Fresnel lens collectors can be used to heat water for space heating or district heating.
- **Cooking**: Fresnel lens collectors can be used for solar cooking.


Parallel light rays
Fresnel lens
Solar cell

Paraboloid dish collector

- A parabolic dish collector (PDC) is a solar heat collector that uses a parabolic-shaped dish to focus sunlight onto a receiver.
- The receiver absorbs the solar energy and converts it into heat or electricity.
- The dish's parabolic shape concentrates sunlight into a single focal point.
- The dish tracks the sun's movement across the sky using a two-axis tracking system.
- The receiver is placed at the focal point of the dish.
- The receiver absorbs the sunlight and converts it into heat or electricity.

- Uses
- PDCs can be used in large-scale power plants.
- PDCs can also be used in small-scale power generation systems for remote facilities.
- PDCs can be used to generate electricity for engines.
-
- Benefits
- PDCs are highly efficient, PDCs have minimal start-up losses, PDCs are modular, and PDCs are cost-effective compared to centralized power generation systems.

Receiver
Reflector
Tracking device
Fluid transfer pipe
Supporting structure

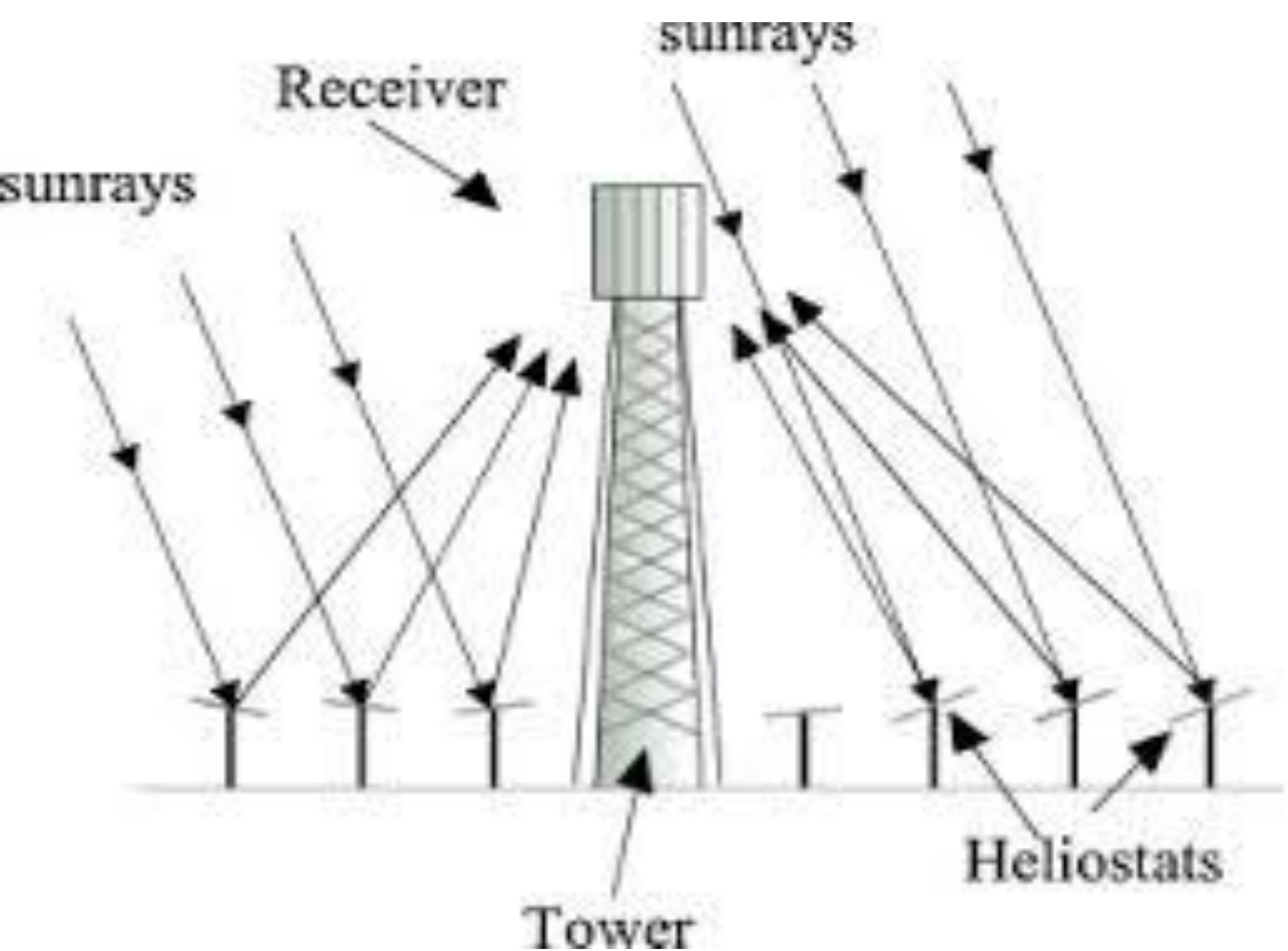
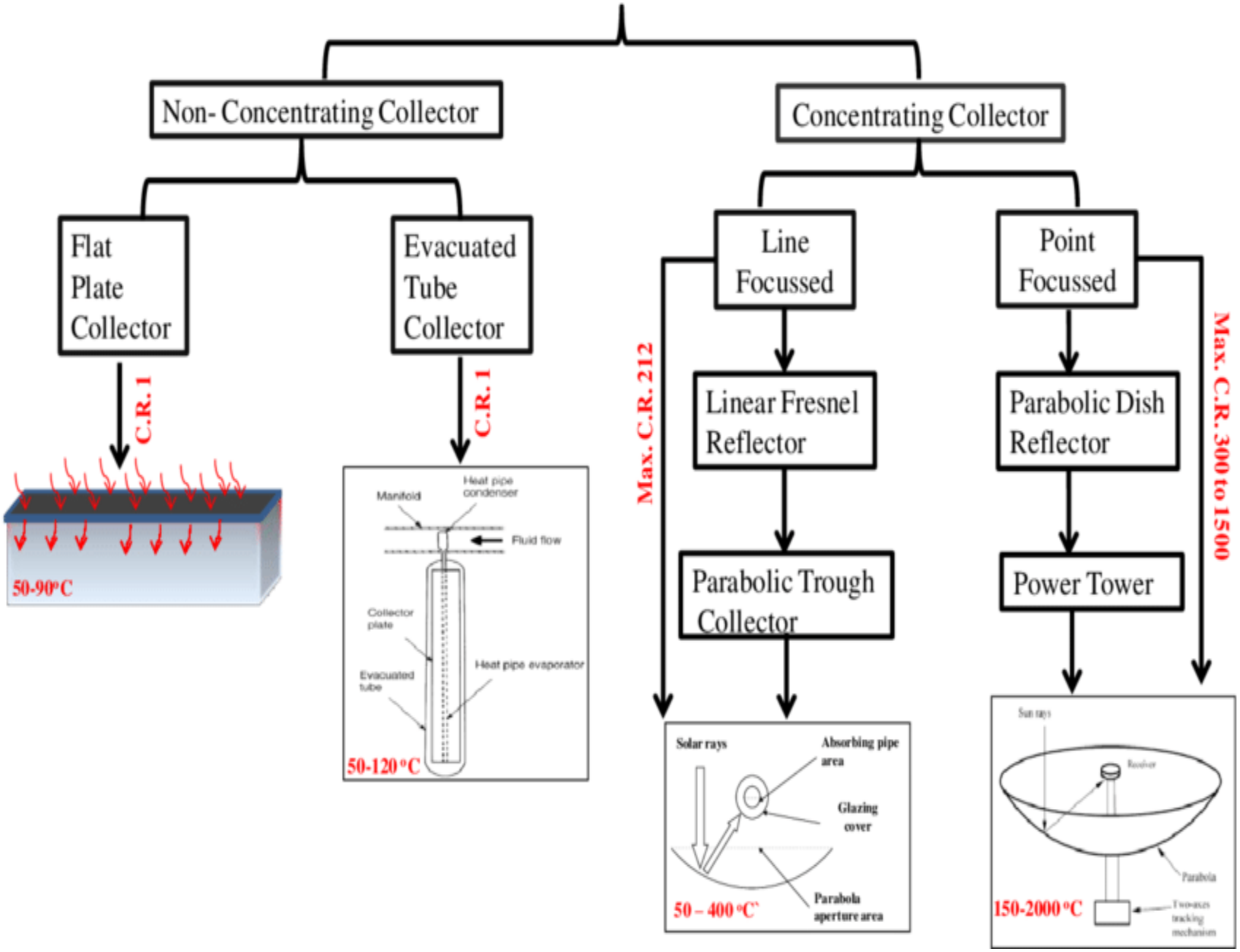
Central Receiver Collector

- A central receiver collector, also known as a heliostat, is a flat mirror that reflects sunlight to a central tower in a solar power tower system.

- How it works
- A field of heliostats track the sun and focus sunlight on a central tower.
- The solar energy is absorbed by a working fluid, such as molten salt or liquid sodium.
- The working fluid is heated and then used to generate electricity.

- Benefits
- Central receiver systems are a viable alternative to fossil fuel power plants.
- They are efficient at collecting solar light and converting it into heat.
- They can be used on non-flat sites.
- They can perform well at large capacities and temperatures.

- Components
- **Heliostats**: Flat mirrors that track the sun and reflect sunlight to the central tower
- **Central receiver**: The tower that receives the sunlight reflected by the heliostats
- **Working fluid**: A fluid that absorbs the solar energy and is used to generate electricity
- **Storage tanks**: Store the heat energy generated by the system
- **Generator**: Uses the heat energy from the working fluid to generate electricity


sunrays
Receiver
sunrays
Heliostats
Tower

Solar Thermal Collectors